Course Code:	Course Title	Credit
CSC601	**System Programming and Compiler Construction**	**3**

Prerequisite: Theoretical computer science, Operating system. Computer Organization and Architecture .	
Course Objectives:	
1	To understand the role and functionality of various system programs over application programs.
2	To understand basic concepts, structure and design of assemblers, macro processors, linkers and loaders.
3	To understand the basic principles of compiler design, its various constituent parts, algorithms and data structures required to be used in the compiler.
4	To understand the need to follow the syntax in writing an application program and to learn how the analysis phase of compiler is designed to understand the programmer ‘s requirements without ambiguity
5	To synthesize the analysis phase outcomes to produce the object code that is efficient in terms of space and execution time
Course Outcomes: On successful completion of course, learner will be able to	
1	Identify the relevance of different system programs.
2	Explain various data structures used for assembler and microprocessor design.
3	Distinguish between different loaders and linkers and their contribution in developing efficient user applications.
4	Understand fundamentals of compiler design and identify the relationships among different phases of the compiler.

Module		Content	Hrs
1		**Introduction to System Software**	**2**
	1.1	Concept of System Software, Goals of system software, system program and system programming, Introduction to various system programs such as Assembler, Macro processor, Loader, Linker, Compiler, Interpreter, Device Drivers, Operating system, Editors, Debuggers.	
2		**Assemblers**	7
	2.1	Elements of Assembly Language programming, Assembly scheme, pass structure of assembler, Assembler Design: Two pass assembler Design and single pass Assembler Design for X86 processor, data structures used.	
3		**Macros and Macro Processor**	6
	3.1	Introduction, Macro definition and call, Features of Macro facility: Simple, parameterized, conditional and nested. Design of Two pass macro processor, data structures used.	
4		**Loaders and Linkers**	**6**
	4.1	Introduction, functions of loaders, Relocation and Linking concept, Different loading schemes: Relocating loader, Direct Linking Loader, Dynamic linking and loading.	
5		**Compilers: Analysis Phase**	**10**
	5.1	Introduction to compilers, Phases of compilers: **Lexical Analysis**- Role of Finite State Automata in Lexical Analysis, Design of Lexical analyzer, data structures used.	

		Syntax Analysis- Role of Context Free Grammar in Syntax analysis, Types of Parsers: Top down parser- LL(1), Bottom up parser- SR Parser, Operator precedence parser, SLR. **Semantic Analysis**, Syntax directed definitions.	
6		**Compilers: Synthesis phase**	**8**
	6.1	**Intermediate Code Generation**: Types of Intermediate codes: Syntax tree, Postfix notation, three address codes: Triples and Quadruples, indirect triple. **Code Optimization**: Need and sources of optimization, Code optimization techniques: Machine Dependent and Machine Independent. **Code Generation:** Issues in the design of code generator, code generation algorithm. Basic block and flow graph.	

Textbooks:	
1	D. M Dhamdhere: *Systems programming and Operating Systems*, Tata McGraw Hill, Revised Second Edition
2	A. V. Aho, R. Shethi, Monica Lam, J.D. Ulman: *Compilers Principles, Techniques and Tools*, Pearson Education, Second Edition.
3	J. J. Donovan: *Systems Programming* Tata McGraw Hill, Edition 1991
References:	
1	John R. Levine, Tony Mason & Doug Brown, *Lex & YACC*, O 'Reilly publication, second Edition
2	D, M .Dhamdhere ,*Compiler construction* 2e, Macmillan publication, second edition .
3	Kenneth C. Louden ,*Compiler construction: principles and practices*, Cengage Learning
4	Leland L. Beck, *System software: An introduction to system programming*, Pearson publication, Third Edition
Useful Links for E-resources:	
1	http://www.nptelvideos.in/2012/11/compiler-design.html
2	https://www.coursera.org/lecture/nand2tetris2/unit-4-1-syntax-analysis-5pC2Z

Assessment:	
Internal Assessment:	
Assessment consists of two class tests of 20 marks each. The first -class test is to be conducted when approx. 40% syllabus is completed and the second-class test when an additional 40% syllabus is completed. Duration of each test shall be one hour.	
End Semester Theory Examination:	
1	Question paper will comprise a total of six questions.
2	All question carries equal marks
3	Questions will be mixed in nature (for example supposed Q.2 has part (a) from module 3 then part (b) will be from any module other than module 3)
4	Only Four questions need to be solved.
5	In question paper weightage of each module will be proportional to number of respective lecture hours as mentioned in the syllabus.

Course Code:	Course Title	Credit
CSC602	**Cryptography & System Security**	**3**

Prerequisite: Computer Networks	
Course Objectives:	
1	To introduce classical encryption techniques and concepts of modular arithmetic and number theory.
2	To explore the working principles and utilities of various cryptographic algorithms including secret key cryptography, hashes and message digests, and public key algorithms
3	To explore the design issues and working principles of various authentication protocols, PKI standards and various secure communication standards including Kerberos, IPsec, and SSL/TLS.
4	To develop the ability to use existing cryptographic utilities to build programs for secure communication
Course Outcomes:	
1	Understand system security goals and concepts, classical encryption techniques and acquire fundamental knowledge on the concepts of modular arithmetic and number theory
2	Understand, compare and apply different encryption and decryption techniques to solve problems related to confidentiality and authentication
3	Apply different message digest and digital signature algorithms to verify integrity and achieve authentication and design secure applications
4	Understand network security basics, analyse different attacks on networks and evaluate the performance of firewalls and security protocols like SSL, IPSec, and PGP
5	Analyse and apply system security concept to recognize malicious code

Module		Content	Hrs
1		**Introduction - Number Theory and Basic Cryptography**	**8**
	1.1	Security Goals, Attacks, Services and Mechanisms, Techniques. Modular Arithmetic: Euclidean Algorithm, Fermat‘s and Euler‘s theorem	
	1.2	Classical Encryption techniques, Symmetric cipher model, mono-alphabetic and polyalphabetic substitution techniques: Vigenere cipher, playfair cipher, Hill cipher, transposition techniques: keyed and keyless transposition ciphers	
2		**Symmetric and Asymmetric key Cryptography and key Management**	**11**
	2.1	Block cipher principles, block cipher modes of operation, DES, Double DES, Triple DES, Advanced Encryption Standard (AES), Stream Ciphers: RC4 algorithm.	
	2.2	Public key cryptography: Principles of public key cryptosystems- The RSA Cryptosystem, The knapsack cryptosystem	
	2.3	Symmetric Key Distribution: KDC, Needham-schroeder protocol. Kerberos: Kerberos Authentication protocol, Symmetric key agreement: Diffie Hellman, Public key Distribution: Digital Certificate: X.509, PKI	
3		**Cryptographic Hash Functions**	**3**
	3.1	Cryptographic hash functions, Properties of secure hash function, MD5, SHA-1, MAC, HMAC, CMAC.	
4		**Authentication Protocols & Digital Signature Schemes**	**5**
	4.1	User Authentication, Entity Authentication: Password Base, Challenge Response Based	

	4.2	Digital Signature, Attacks on Digital Signature, Digital Signature Scheme: RSA	
5		**Network Security and Applications**	**9**
	5.1	Network security basics: TCP/IP vulnerabilities (Layer wise), Network Attacks: Packet Sniffing, ARP spoofing, port scanning, IP spoofing	
	5.2	Denial of Service: DOS attacks, ICMP flood, SYN flood, UDP flood, Distributed Denial of Service	
	5.3	Internet Security Protocols: PGP, SSL, IPSEC. Network security: IDS, Firewalls	
6		**System Security**	**3**
	6.1	Buffer Overflow, malicious Programs: Worms and Viruses, SQL injection	

Textbooks:	
1	William Stallings, ***"Cryptography and Network Security, Principles and Practice"***, 6th Edition, Pearson Education, March 2013
2	Behrouz A. Ferouzan, ***"Cryptography & Network Security"***, Tata McGraw Hill
3	Behrouz A. Forouzan & Debdeep Mukhopadhyay, ***"Cryptography and Network Security"*** 3rd Edition, McGraw Hill

Referecebooks:	
1	Bruce Schneier, ***"Applied Cryptography, Protocols Algorithms and Source Code in C",*** Second Edition, Wiley.
2	Atul Kahate, ***"Cryptography and Network Security",*** Tata McGraw-Hill Education, 2003.
3	Eric Cole, ***"Network Security Bible",*** Second Edition, Wiley, 2011.

Assessment:	
Internal Assessment:	
Assessment consists of two class tests of 20 marks each. The first class test is to be conducted when approx. 40% syllabus is completed and second class test when additional 40% syllabus is completed. Duration of each test shall be one hour.	
End Semester Theory Examination:	
1	Question paper will comprise of total six questions.
2	All question carries equal marks
3	Questions will be mixed in nature (for example supposed Q.2 has part (a) from module 3 then part (b) will be from any module other than module 3)
4	Only Four question need to be solved.
5	In question paper weightage of each module will be proportional to number of respective lecture hours as mention in the syllabus.

Useful Links	
1	https://github.com/cmin764/cmiN/blob/master/FII/L3/SI/book/W.Stallings%20-%20Cryptography%20and%20Network%20Security%206th%20ed.pdf
2	https://docs.google.com/file/d/0B5F6yMKYDUbrYXE4X1ZCUHpLNnc/view

Course Code:	Course Title	Credit
CSC603	**Mobile Computing**	**3**

Prerequisite: Computer Networks	
Course Objectives:	
1	To introduce the basic concepts and principles in mobile computing. This includes major techniques involved, and networks & systems issues for the design and implementation of mobile computing systems and applications.
2	To explore both theoretical and practical issues of mobile computing.
3	To provide an opportunity for students to understand the key components and technologies involved and to gain hands-on experiences in building mobile applications.
Course Outcomes: On successful completion of course, learner will be able to	
1	To identify basic concepts and principles in computing, cellular architecture.
2	To describe the components and functioning of mobile networking.
3	To classify variety of security techniques in mobile network.
4	To apply the concepts of WLAN for local as well as remote applications.
5	To describe Long Term Evolution (LTE) architecture and its interfaces.

Module		Content	Hrs
1		**Introduction to Mobile Computing**	**4**
	1.1	Introduction to Mobile Computing, Telecommunication Generations, Cellular systems,	
	1.2	Electromagnetic Spectrum, Antenna, Signal Propagation, Signal Characteristics, Multiplexing, Spread Spectrum: DSSS & FHSS, Co-channel interference	
2		**GSM Mobile services**	**8**
	2.1	GSM Mobile services, System Architecture, Radio interface, Protocols, Localization and Calling, Handover, security (A3, A5 & A8)	
	2.2	GPRS system and protocol architecture	
	2.3	UTRAN, UMTS core network; Improvements on Core Network,	
3		**Mobile Networking**	**8**
	3.1	Medium Access Protocol, Internet Protocol and Transport layer	
	3.2	Mobile IP: IP Packet Delivery, Agent Advertisement and Discovery, Registration, Tunneling and Encapsulation, Reverse Tunneling.	
	3.3	Mobile TCP: Traditional TCP, Classical TCP Improvements like Indirect TCP, Snooping TCP & Mobile TCP, Fast Retransmit/ Fast Recovery, Transmission/Timeout Freezing, Selective Retransmission	
4		**Wireless Local Area Networks**	**6**
	4.1	**Wireless Local Area Networks:** Introduction, Infrastructure and ad-hoc network	
	4.2	**IEEE 802.11:**System architecture , Protocol architecture , Physical layer, Medium access control layer, MAC management, 802.11a, 802.11b standard	
	4.3	Wi-Fi security : WEP ,WPA, Wireless LAN Threats , Securing Wireless Networks	

	4.4	Bluetooth: Introduction, User Scenario, Architecture, protocol stack	
5		**Mobility Management**	**6**
	5.1	**Mobility Management :** Introduction, IP Mobility, Optimization, IPv6	
	5.2	Macro Mobility : MIPv6, FMIPv6	
	5.3	Micro Mobility: CellularIP, HAWAII, HMIPv6	
6		**Long-Term Evolution (LTE) of 3GPP**	**7**
	6.1	**Long-Term Evolution (LTE) of 3GPP :** LTE System Overview, Evolution from UMTS to LTE	
	6.2	LTE/SAE Requirements, SAE Architecture	
	6.3	EPS: Evolved Packet System, E-UTRAN, Voice over LTE (VoLTE), Introduction to LTE-Advanced	
	6.4	Self Organizing Network (SON-LTE), SON for Heterogeneous Networks (HetNet), Comparison between Different Generations (2G, 3G, 4G and 5G), Introduction to 5G	

Textbooks:	
1	Jochen Schilller, "**Mobile Communication**", Addision wisely, Pearson Education
2	William Stallings "**Wireless Communications & Networks**", Second Edition, Pearson Education
3	Christopher Cox, "**An Introduction to LTE: LTE, LTE-Advanced, SAE and 4G Mobile Communications**", Wiley publications
4	Raj Kamal, **"Mobile Computing"**, 2/e, Oxford University Press-New
References:	
1	Seppo Hamalainen, Henning Sanneck , Cinzia Sartori, **"LTE Self-Organizing Networks (SON): Network Management Automation for Operational Efficiency",** Wiley publications
2	Ashutosh Dutta, Henning Schulzrinne **"Mobility Protocols and Handover Optimization: Design, Evaluation and Application",** IEEE Press, Wiley Publication
3	Michael Gregg, **"Build your own security lab",** Wiley India edition
4	Dipankar Raychaudhuri, Mario Gerla, **"Emerging Wireless Technologies and the Future Mobile Internet",** Cambridge
5	Andreas F. Molisch, **"Wireless Communications",** Second Edition, Wiley Publication

Assessment:	
Internal Assessment:	
Assessment consists of two class tests of 20 marks each. The first class test is to be conducted when approx. 40% syllabus is completed and second class test when additional 40% syllabus is completed. Duration of each test shall be one hour.	
End Semester Theory Examination:	
1	Question paper will comprise of total six questions.
2	All question carries equal marks
3	Questions will be mixed in nature (for example supposed Q.2 has part (a) from module 3 then part (b) will be from any module other than module 3)
4	Only Four question need to be solved.
5	In question paper weightage of each module will be proportional to number of respective lecture hours as mention in the syllabus.

Course Code:	Course Title	Credit
CSC604	**Artificial Intelligence**	**3**

Prerequisite: Discrete Mathematics, Data Structures	
Course Objectives:	
1	To conceptualize the basic ideas and techniques underlying the design of intelligent systems.
2	To make students understand and Explore the mechanism of mind that enables intelligent thought and action.
3	To make students understand advanced representation formalism and search techniques.
4	To make students understand how to deal with uncertain and incomplete information.
Course Outcomes: At the end of the course, the students will be able to	
1	Ability to develop a basic understanding of AI building blocks presented in intelligent agents.
2	Ability to choose an appropriate problem solving method and knowledge representation technique.
3	Ability to analyze the strength and weaknesses of AI approaches to knowledge– intensive problem solving.
4	Ability to design models for reasoning with uncertainty as well as the use of unreliable information.
5	Ability to design and develop AI applications in real world scenarios.

Module		Content	Hrs
1		**Introduction to Artificial Intelligence**	**4**
	1.1	Introduction, History of Artificial Intelligence, Intelligent Systems: Categorization of Intelligent System, Components of AI Program, Foundations of AI, Sub-areas of AI, Applications of AI, Current trends in AI.	
2		**Intelligent Agents**	**4**
	2.1	Agents and Environments, The concept of rationality, The nature of environment, The structure of Agents, Types of Agents, Learning Agent.	
	2.2	Solving problem by Searching: Problem Solving Agent, Formulating Problems, Example Problems.	
3		**Problem solving**	**10**
	3.1	Uninformed Search Methods: Breadth First Search (BFS), Depth First Search (DFS), Depth Limited Search, Depth First Iterative Deepening (DFID), Informed Search Methods: Greedy best first Search, A* Search, Memory bounded heuristic Search.	
	3.2	Local Search Algorithms and Optimization Problems: Hill climbing search Simulated annealing, Genetic algorithms.	
	3.3	**Adversarial Search:** Game Playing, Min-Max Search, Alpha Beta Pruning	
4		**Knowledge and Reasoning**	**12**
	4.1	Knowledge based Agents, Brief Overview of propositional logic, First Order Logic: Syntax and Semantic, Inference in FOL, Forward chaining, backward Chaining.	
	4.2	Knowledge Engineering in First-Order Logic, Unification, Resolution	

	4.3	Uncertain Knowledge and Reasoning: Uncertainty, Representing knowledge in an uncertain domain, The semantics of belief network, Simple Inference in belief network	
5		**Planning and Learning**	**5**
	5.1	The planning problem, Planning with state space search, Partial order planning, Hierarchical planning, Conditional Planning.	
	5.2	Learning: Forms of Learning, Theory of Learning, PAC learning. Introduction to statistical learning (Introduction only) Introduction to reinforcement learning: Learning from Rewards, Passive Reinforcement Learning, Active reinforcement Learning	
6		**AI Applications**	**4**
		A. Introduction to NLP- Language models, Grammars, Parsing B. Robotics - Robots, Robot hardware, Problems Robotics can solve C. AI applications in Healthcare, Retail, Banking	

Textbooks:	
1	Stuart J. Russell and Peter Norvig, "***Artificial Intelligence: A Modern Approach***", Fourth Edition" Pearson Education, 2020.
2	Saroj Kaushik, "***Artificial Intelligence***", Cengage Learning, First edition, 2011
3	George F Luger, "***Artificial Intelligence***" Low Price Edition, Fourth edition, Pearson Education.,2005
References:	
1	Nils J. Nilsson, Principles of Artificial Intelligence, Narosa Publication.
2	Deepak Khemani, A First Course in Artificial Intelligence, McGraw Hill Publication
3	Patrick H. Winston, Artificial Intelligence, 3rd edition, Pearson Education.
4	Elaine Rich and Kevin Knight, "***Artificial Intelligence***", Third Edition, McGraw Hill Education,2017.

Assessment:	
Internal Assessment:	
Assessment consists of two class tests of 20 marks each. The first class test is to be conducted when approx. 40% syllabus is completed and the second class test when an additional 40% syllabus is completed. Duration of each test shall be one hour.	
End Semester Theory Examination:	
1	Question paper will comprise a total of six questions.
2	All question carries equal marks
3	Questions will be mixed in nature (for example supposed Q.2 has part (a) from module 3 then part (b) will be from any module other than module 3)
4	Only Four questions need to be solved.
5	In question paper weightage of each module will be proportional to number of respective lecture hours as mentioned in the syllabus.

Useful Links	
1	https://nptel.ac.in/courses/106/105/106105078/
2	https://thestempedia.com/blog/simple-ai-and-machine-learning-projects-for-students-and-beginners/
3	https://nptel.ac.in/courses/106/105/106105079/

Course Code:	Course Title	Credit
CSDLO6013	**Quantitative Analysis**	**3**

Prerequisite: Applied Mathematics	
Course Objectives:	
1	Introduction to the basic concepts in Statistics
2	Understand concept of data collection & sampling methods.
3	Introduction to Regression, Multiple Linear Regression
4	Draw interference using Statistical inference methods
5	Tests of hypotheses
Course Outcomes:	
1	Recognize the need of Statistics and Quantitative Analysis
2	Apply the data collection and the sampling methods.
3	Analyze using concepts of Regression, Multiple Linear Regression
4	Formulate Statistical inference drawing methods.
5	Apply Testing of hypotheses

Module	Content	Hrs
1	**Introduction to Statistics**	**6**
	Functions – Importance – Uses and Limitations of Statistics. Statistical data– Classification, Tabulation, Diagrammatic & Graphic representation of data	
2	**Data Collection & Sampling Methods**	**6**
	Primary & Secondary data, Sources of data, Methods of collecting data. Sampling – Census & Sample methods –Methods of sampling, Probability Sampling and Non-Probability Sampling.	
3	**Introduction to Regression**	**8**
	Mathematical and Statistical Equation – Meaning of Intercept and Slope – Error term – Measure for Model Fit –R2 – MAE – MAPE.	
4	**Introduction to Multiple Linear Regression**	**8**
	Multiple Linear Regression Model, Partial Regression Coefficients, Testing Significance overall significance of Overall fit of the model, Testing for Individual Regression Coefficients	
5	**Statistical inference**	**6**
	Random sample -Parametric point estimation unbiasedness and consistence - method of moments and method of maximum likelihood.	
6	**Tests of hypotheses**	**5**
	Null and Alternative hypotheses. Types of errors. Neyman-Pearson lemma-MP and UMP tests.	

Textbooks:	
1	Agarwal, B.L. (2006):-Basic Statistics. Wiley Eastern Ltd., New Delhi
2	Gupta, S. P. (2011):-Statistical Methods. Sultanchand&Sons, New Delhi
3	Sivathanupillai, M &Rajagopal, K. R. (1979):-Statistics for Economics Students.
4	Hogg ,R.V. and Craig, A.T.(2006), An introduction to mathematical statistics, Amerind publications.
References:	